Safety of sodium-glucose cotransporter 2 inhibitors on amputation across categories of baseline cardiovascular disease and diuretics in patients with type 2 diabetes

Running Title: Safety of SGLT2i on Amputation by CVD and Diuretics

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Abstract

Aims: To assess the risk of amputation associated with sodium-glucose cotransporter 2 inhibitors (SGLT2is) among patients with type 2 diabetes, across categories of baseline cardiovascular disease (CVD) and diuretic use (DU).

Materials and Methods: We conducted an active comparator, new-user cohort study using Korea's nationwide claims data (2015–2020). Study cohort consisted of patients with type 2 diabetes who initiated SGLT2is or dipeptidyl peptidase-4 inhibitors (DPP4is). Cohort entry was defined by first prescription date. We then classified patients into four discrete subcohorts based on their baseline status of CVD and DU as: (1) CVD+/DU+, (2) CVD+/DU-, (3) CVD-/DU+, and (4) CVD-/DU-. We performed 1:1 propensity score (PS) matching within each cohort and estimated hazard ratios (HRs) with 95% confidence intervals (CIs) for the risk of amputation with SGLT2is versus DPP4is using Cox models.

Results: We identified 219,900 PS-matched pairs of SGLT2i and DPP4is (CVD+/DU+, n=11,719; CVD+/DU-, n=26,092; CVD-/DU+, n=26,894; and CVD-/DU-, n=155,195), with well-balanced baseline covariates across all cohorts. Significantly lower risks of amputation with SGLT2is versus DPP4is were found in CVD+/DU+ (HR 0.36, 95% CI 0.14-0.90), CVD+/DU- (0.45, 0.21-0.99), and CVD-/DU- (0.48, 0.33-0.70), but not in CVD-/DU+ (0.54, 0.26-1.12). Consistent trends in estimates were found across various sensitivity analyses.

Conclusions: Initiating SGLT2is against DPP4is did not increase the risk of amputation across patient populations of varying vulnerability. These findings based on routine practice reassure clinicians of the safety of SGLT2is with regard to amputation risks in select high-risk patients with type 2 diabetes.

INTRODUCTION

Sodium-glucose co-transporter 2 inhibitors (SGLT2is), the most recently available class of oral antidiabetic agents, have been shown to have benefits beyond glucose lowering.[1] Given the cardiorenal benefits previously established by several landmark trials and the subsequent emphasis in recent clinical guidelines for the management of type 2 diabetes (T2D),[1] the use of SGLT2is has been substantially increasing in patients with cardiovascular disease (CVD).

Despite these benefits, SGLT2is have been not free from potential risks of amputation since the excess major and minor amputation risks with canagliflozin was unexpectedly detected in CANVAS trial.[2 3] In contrast, CREDENCE trial reported no increased risk of amputation with canagliflozin in diabetic patients with comorbid chronic renal disease (HR 1.11, 95% CI 0.79-1.56).[4] The incidence of amputation was also not significantly increased in cardiovascular outcome trials with dapagliflozin and empagliflozin.[5 6] However, heterogeneity between study population in trials has made difficult to conclude whether SGLT2is are safe for amputation risk: target population of at cardiovascular risk or comorbid renal disease,[2-4 7] or difference in prevalence of atherosclerotic disease.[2 5] The concern with trials and its meta-analysis was also the low number of events, associated with a high level of uncertainty, even if pooling all subtypes of amputations together,[8] and the possibility of underestimation of minor amputations remain in trials.[9 10]

Findings were inconsistent from several observational studies also depending on the study population characteristics.[11-17] A recent study reported an increased amputation risk with canagliflozin among patients aged ≥65 years with baseline CVD (HR 1.73, 95% CI 1.30-2.29) but not for those aged <65 years without baseline CVD (HR 1.09, 95% CI 0.83-1.43) [12]. Another study reported a four-fold elevated amputation risk with SGLT2is in patients with peripheral artery disease, as compared with other antidiabetic drugs [17]. Despite

summarised evidence of trials and observational studies suggesting safety of SGLT2is on amputation compared to placebo[18-20] and DPP4i,[21 22] potential amputation risk with SGLT2i cannot be yet ruled out in high-risk patients based on existing data and biological hypothesis.[23]

Given this evidence gap and recent indication expansion to heart failure of SGLT2is,[24 25] there is a need to generate further evidence to date on this clinical issue in patients with strong risk factors of amputation; particularly in patients with CVD[26 27] and be likely treated with diuretics which have a shared mechanism with SGLT2is for amputation risk.[23 28 29] It is important to contextualise the potential amputation risks with SGLT2is across categories of background CVD and/or diuretic use (DU) in patients to help prevent potentially avoidable amputations. Therefore, this study focuses on the risk of amputation with SGLT2is in each patient population using dipeptidyl peptidase-4 inhibitors (DPP4is) as an active comparator, which is the most commonly used second-line therapy for T2D, and also has been reassured about safety on amputation risk and excellent safety profile by previous studies.[30]

MATERIALS AND METHODS

Data Source

We used the Health Insurance Review & Assessment (HIRA) research database (Data No. M20210607316), a nationwide healthcare claims data of South Korea. Under the term of universal health coverage, the insurance system covers approximately 50 million (about 98% of the total population) as of 2014. HIRA database, generated in the process of reimbursement, contains longitudinal, patient-level, anonymized patient data on diagnoses, surgical and medical procedures, prescription drug claims from all care settings.[31] Records of diagnoses are coded based on the International Classification of Diseases, 10th Revision

(ICD-10), and overall positive predictive value (PPV) for diagnostic code in HIRA database was 82% when comparing electronic medical records in validation study.[32] The database is widely used resource for observational descriptive and clinical research.[33-35]

The study protocol was approved by the institutional review board of Sungkyunkwan University (IRB No. SKKU2021-08-004). As we used anonymized retrospective data, the board waived the requirement for obtaining informed consent.

Study Population

We identified patients with T2D between 2016 and 2020 who were newly prescribed drug of interest or comparator drugs: SGLT2is (dapagliflozin, empagliflozin, ertugliflozin, or ipragliflozin; canagliflozin is not available in South Korea) or DPP4is. T2D was defined by ICD-10 codes (E11-14) in all care setting. Cohort entry was defined as the date of the first prescription for SGLT2is or DPP4is, alone or in combination with other antidiabetic agents excluding with one another. We defined new users as patients without any previous prescription for these drugs within the year before cohort entry. We then excluded patients (A) aged <18 years at cohort entry, (B) initiated both SGLT2is and DPP4is at cohort entry, (C) had end-stage renal disease or dialysis (i.e., contraindications to SGLT2is) within the year before cohort entry, and (D) had a history of amputation at any time before or at cohort entry (**Figure S1**).

We then classified all eligible patients into four mutually exclusive groups based on the baseline status, assessed within the year before cohort entry, of CVD and/or DU as: (A) CVD+/DU+, patients with CVD and DU; (B) CVD+/DU-, patients with CVD but no DU; (C) CVD-/DU+, patients without CVD but DU; (D) CVD-/DU-, patients without CVD and no DU. Patients with CVD were defined as having ≥1 diagnosis of cerebrovascular disease, coronary artery disease, heart failure, or peripheral vascular disease at baseline, and DU was

defined as patients with ≥1 prescription of loop diuretics, thiazide or thiazide-like diuretics, mineralocorticoid receptor antagonists, or other diuretics at baseline assessment period, within a year before cohort entry. Detailed diagnostic, anatomical therapeutic chemical, or procedure codes used to define the study cohort and each subgroup are shown in **Table S1.**

Exposure and Follow-up

Exposure was defined using an as-treated approach based on the cohort entry drug, with patients followed from cohort entry until outcome occurrence, treatment discontinuation (gap of ≥30 days in between successive prescriptions), initiation of a DPP4i among SGLT2i users or vice versa, death, or end of study period, whichever occurred first.

Outcome

Outcome of interest was incident nontraumatic limb amputation, defined as amputation of extremities using procedure codes documented in physician-based claims data. The procedure codes used to define amputations are reported in **Table S1**. To differentiate and exclude amputations from other causes (e.g., trauma), we considered our outcome of interest as those that did not accompany ICD-10 diagnostic codes of "S" or "T" in either the primary or secondary position.

Propensity Score Matching

We used propensity score (PS) matching to adjust potential confounding derived from differences in patients prescribed SGLT2is versus DPP4is.[36 37] PS was estimated using a multivariable logistic regression model that included all potential confounders assessed at cohort entry or during the baseline period, a year before cohort entry: age, sex, healthcare utilization, diabetes-related conditions, proxy for diabetes severity, other comorbidities and

use of comedications related to amputation;[37] the comprehensive list and detailed descriptions are listed in **Table S2**. PS was estimated in each patient population separately. After PS estimation, we performed 1:1 nearest neighbour matching without replacement with a caliper width of 0.05 on the PS scale.

Statistical Analyses

Baseline patient characteristics were summarized individually for each cohort using frequencies and percentages for categorical variables, and mean and standard deviation (SD) for continuous variables. We assessed covariate balance between treatment groups using the absolute standardized mean difference (<0.1 indicates negligible differences) before and after PS matching.[36] We estimated the rate difference per 1000 person-years assuming the Poisson distribution, and hazard ratios (HRs) with 95% CIs using the Cox proportional hazards regression models within the PS-matched cohort without further adjustment. Before the survival analysis, the Cox proportional hazards assumptions were evaluated using time interaction term. To show the cumulative proportion of amputation over time, we plotted the cumulative incidences of amputations among SGLT2i and DPP4i groups in each cohort, and the significance of difference was assessed using Gray's test. We also performed age- and sex-stratified analyses to investigate the effect modification of age (<65 vs. ≥65 years) and sex. Interaction was tested using a Wald test for homogeneity.[38]

We performed several sensitivity analyses. First, we repeated the primary analysis using gaps of 0 and 60 days to define continuation of study drugs. Second, we conducted intention-to-treat analysis. Exposure was defined at cohort entry and patients were followed until the occurrence of outcome or censored due to death, 365th day of follow-up, or end of the study period, whichever occurred first; treatment discontinuation and switching after the start of follow-up were disregarded in this setting. Third, alternative definitions of baseline

CVD and DU were implemented, whereby ≥2 records of diagnosis or prescriptions were required instead of ≥1. Fourth, we applied alternative definition for new users of study drugs with a washout period of two years. Fifth, as a positive control outcome, we assessed the risk of hospitalization for heart failure with SGLT2is comparing DPP4is to evaluate the reproducibility for a known association and determine the internal validity of our data; previous studies have shown a lower risk of hospitalization for heart failure with SGLT2is.[19] Last, a post-hoc sensitivity analysis using the E-value was conducted to assess the how strong a potential effect of unmeasured or unaccounted residual confounding would have to be to disregard an observed association.[39]

As an auxiliary analysis, we repeated our primary analysis in the overall study cohort to examine the risk of amputation with SGLT2is regardless of baseline CVD and DU; PS were re-estimated in the overall cohort.

All analyses were conducted using the SAS Enterprise Guide version 7.1 (SAS Institute Inc., Cary, NC, USA).

RESULTS

A total of 1,610,006 patients with T2D were newly prescribed SGLT2is or DPP4is in South Korea between 2016 and 2020. After applying all eligibility criteria, we identified 1,588,797 patients before PS matching (57.8% male; mean [SD] age, 60.2 [13.7] years), of which, 219,976 (13.8%) were prescribed SGLT2is (**Figure S2**). Before PS matching, the proportions of each cohort were 6.6% (CVD+/DU+), 13.4% (CVD+/DU-), 13.6% (CVD-/DU+), and 66.7% (CVD-/DU-). All baseline covariates achieved balance after PS matching, with absolute standardized mean differences <0.1 (**Table 1**), indicating similar distributions of these variables between treatment groups, and the effect estimates could be interpreted as association between exposure and outcome independent of measured covariates.

Comprehensive baseline characteristics appear in **Table S3** to **Table S7** for each stratified and overall cohort, before and after PS matching. On the whole characteristics after PS matching across cohorts, cohort of CVD+ or DU+ tended to be older, have higher proportions of various comorbidity and comedications, and have higher level of antidiabetic treatment and healthcare utilization. Additionally, about 50% of patients in cohorts of DU+ (CVD+/DU+ and CVD-/DU+) are female, while about 40% in DU- (CVD-/DU- and CVD-/DU-).

The SGLT2is users had a lower rate of amputations than DPP4is across all cohorts after PS matching. In general, cohorts of CVD+ or DU+ had a higher rate of amputation per 1000 person-years than their counterpart cohorts, with observed rates as follows, respectively: 0.45 vs 1.21 (CVD+/DU+); 0.30 vs 0.64 (CVD+/DU-); 0.31 vs 0.57 (CVD-/DU+); 0.25 vs 0.51 (CVD-/DU-); 0.28 vs 0.50 (overall cohort) (**Table 2**). The cumulative incidence curves for amputation in each stratified cohort appear in **Figure 1** and the overall cohort in **Figure S3**. Absolute risks of amputation were lower in SGLT2is than DPP4is in all cohorts with corresponding rate differences of -0.76 per 1000 person-years (-1.42 to -0.99) in CVD+/DU+, -0.34 (-0.68 to -0.00) in CVD+/DU-, -0.26 (-0.39 to -0.13) in CVD-/DU-, and -0.22 (0.33 to -0.11) in overall cohort (**Figure 2A**). The PS-matched HRs for amputation associated with SGLT2is comparing DPP4is were: 0.36 (95% CI 0.14-0.90) in CVD+/DU+, 0.45 (0.21-0.99) in CVD+/DU-, 0.54 (0.26-1.12) in CVD-/DU+, 0.48 (0.33-0.70) in CVD-/DU-, and 0.54 (0.40 to 0.73) in overall cohort (**Figure 2B**); results of both absolute and relative risks of amputation associated with SGLT2is are shown in **Table S8**.

No significant interactions were found with age or sex on amputation risks across all stratified cohorts (**Table S8**). Results of all sensitivity analyses were in line with the main analysis (**Table S9**). The risk of hospitalization for heart failure, a positive control outcome, was lower with SGLT2is than DPP4is (**Table S10**). The E-values for point estimates in the

CVD+/DU+, CVD+/DU-, CVD-/DU+, and CVD-/DU- cohorts were 5.00, 3.87, 3.11, and 3.59, respectively.

DISCUSSION

This cohort study using nationwide claims data from South Korea found a generally lower risk of amputation with SGLT2is compared with DPP4is in all categories of baseline CVD and DU in routine clinical practice. These findings suggest that use of SGLT2is did not increase the risk of amputation in high-risk patients (CVD+/DU+) as well as in patients with relatively lower amputation risks (CVD-/DU-). Notably, the point effect estimate for amputation was lowest with SGLT2is in patients at highest risk of amputations (CVD+/DU+).

In addition to ensuring the safety of SGLT2is on amputation risk, our findings could potentially suggest the possibility of favourable effect of SGLT2is to prevent amputation events in specific patient population. The direction and magnitude of effect estimates of SGLT2is obtained across the context of patients at high-risk indicates that the more patient has factors of CVD and DU, the lower amputation risk with SGLT2i compared to DPP4i (**Table 2** and **Figure 2**). Potential explanation for this paradox result could be cardiovascular and renal protection effect of SGLT2is established in various previous studies.[40-42] As for the patients with DU, the increased risk of amputation could be potentially related to renal impairment from the indication for diuretics apart from its mechanisms of action. Together with cardiovascular impairment in CVD group, our findings could be interpreted that harmful effect of cardiorenal deterioration on amputation might be attenuated by the effect of SGLT2is. Further research in other population and on the exact biological mechanisms are needed to determine this potential unexpected benefit of SGLT2is.

Since questions arose regarding amputation risks with SGLT2is, a large body of studies has been published investigating this association concern all SGLT2is beyond

canagliflozin.[8 21] In contrast to the accumulated evidence in patients with T2D in general, very few studies have specifically assessed this link in high-risk patient groups. With CVD and DU known to be two strong risk factors of low limb flow and subsequent amputation, incidence rates of amputation were different across the fore mentioned patient groups in this study as anticipated. The rate of amputation was highest in CVD+/DU+ (0.45 and 1.21 per 1000 person-years for SGLT2i and DPP4i), or patients at high risk, which was a rate that was almost 2-fold higher than that in CVD-/DU- (0.25 and 0.51 per 1000 person-years) (**Table 2**). Moreover, a stronger association between SGLT2is and lower amputation risks was observed with higher underlying risks. Our findings, although indirect, were also consistent with a post-hoc analysis on a subgroup of the CANVAS trial as canagliflozin demonstrated greater benefits in patients on baseline DU; adverse events related to volume depletion or osmotic diuresis were not increased, in addition to the greater cardiovascular benefits.[43] One observational study did report conflicting results of a higher risk of amputation with canagliflozin in patients aged ≥65 years with baseline CVD (HR 1.73, 95% CI 1.30-2.29), when having investigated this association according to age and CVD.[12] However, important challenges exist when directly comparing these findings with ours as canagliflozin was not available in Korea.

We found a generally lower rate of amputation than those of previous studies. Results from the auxiliary analysis showed that the rates of amputation in the overall PS-matched cohort were 0.28 and 0.50 per 1000 person-years among users of SGLT2is and DPP4is, which is considerably lower than a recent meta-analysis of observational studies with 2.48 and 2.67 per 1000 person-years, respectively.[21] However, this is compatible with evidence from previous Korean studies on amputation although it is hard to compare directly the amputation rates due to the different study population: patients win diabetes,[44] and diabetic foot ulcer.[45] The differences in amputation rates may be explained by ethnicity. Most

previous studies included in aforementioned meta-analysis included European and American populations,[21] where several data have previously suggested racial and ethnic differences in the risk of amputations, with particularly higher rates of these outcomes among Europeans and Americans than other races, including Asians.[46-48] Another potential reason could be high-quality primary care in Korea; the data provided by OECD library statistics on amputation rates showed that the international variation is 18-fold and Korea was one of the countries with lowest rates of amputation.[49] Given that little data on this space have come from Asian populations, our findings contribute to the existing literature and help fill any knowledge gaps.

It is important to contextualize the risk of amputation among high-risk patients in T2D management, because amputation is uncommon but serious complication of diabetes carrying subsequent increased risk of morbidity and mortality, and also be potentially avoidable event.[50] In light of the growing use of SGLT2is in patients with T2D and comorbid heart failure, who commonly receive diuretics, our findings could help clinicians, guideline writers, and patients in their understanding of the risk of amputations and preventing avoidable amputation event in routine clinical practice.

Our study has several limitations. First, despite attempts to adjust for potential confounding through the use of an active comparator new-user design and PS methods, residual unmeasured confounding cannot be ruled out. Second, given the hypothesis regarding amputation risk with SGLT2is, our operational definition of DU group is potentially problematic. As diuretics have a shared mechanism of action with SGLT2is on amputation risk and also have an increased risk of amputation under its use, the effect of SGLT2is in DU group which we aimed to investigate was related to taking diuretics. However, our operational definition of DU group reflects patients who potentially use diuretics after baseline period assuming chronic use of diuretics. Third, there is potential

exposure misclassification of the exposure due to the inherent nature of observational studies using secondary data, as we were unable to know adherence to the prescribed drug. Presence of the prescription does not indicate actual medication intake, and the absence of a prescription does not rule out medication use; this discrepancy could introduce potential biases. However, results of sensitivity analyses applying the intention-to-treat definition and varying grace period to define treatment continuation showed consistent findings to that of the main results. Based on the results from sensitivity analysis (**Table S9**), we could anticipate that any misclassification from inaccurate adherence would unlikely to significantly affect our effect estimates. Fourth, despite the attempts to minimize any outcome misclassification by using highly specific procedure codes to detect amputation, we cannot completely exclude the possibility of underestimation or misclassification of outcome. Early detection of signs and symptoms that may alert the clinician to the possibility of amputation and consequent medical management, including discontinuation of SGLT2is, could be possible before the decision of amputation. Therefore, our findings should be interpreted with caution with the potential bias from informative censoring. Fifth, although prior amputation is one of the major risk factors for amputation,[3] we could not assess the recurrent amputation with SGLT2is because we exclude patients with history of amputation to focus our analysis on incident cases of amputation and to minimize potential confounding related to prior amputations. Last, we could not investigate the effect of SGLT2is in each patient group on specific subtypes of amputation, especially major and minor amputations. Considering the potential difference between major and minor amputation associated with pathophysiology, further studies are needed to explore the association of subtypes of amputation with SGLT2is in various patient groups.

Among patients with T2D, the use of SGLT2is compared with DPP4is was not associated with an increased risk of amputations across patient groups according to baseline

CVD and/or DU. Given the anticipated increasing use of SGLT2is in patients with T2D particularly in comorbid heart failure, with the latter likely at a high-risk of amputation, findings from this study reassure clinicians on the safety of SGLT2is regarding amputation across various patient populations subject to different underlying risk of this severe complication.

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Ethics Approval and Consent to Participate: This study has been performed in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of the Sungkyunkwan University approved this study protocol (IRB No. SKKU2021-08-004). The Institutional Review Board of the Sungkyunkwan University also approved that informed consent was not required as this study used anonymized administrative data; this board waived the requirement for obtaining informed consent from all patients.

Consent to Publish: Not Applicable

Data Availability Statement: The data that support the findings of this study are available from the Health Insurance Review and Assessment Service of South Korea but restrictions apply to the availability of these data due to domestic laws and regulations that prohibit the distribution or release of individual's data to the public, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of the Health Insurance Review and Assessment Service of South Korea.

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Legends to figures

Figure 1. Cumulative incidence curves after propensity score matching for each of the four groups stratified by baseline cardiovascular disease and diuretic use.

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

Abbreviations: CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitors; DU, diuretic; SGLT2i, sodium-glucose cotransporter 2 inhibitor

Figure 2. Forest plot for propensity score matched rate differences and hazard ratios of amputation for each of the four groups stratified by baseline cardiovascular disease and diuretic use and overall cohort.

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

Abbreviations: CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; PY, person years; SGLT2i, sodium-glucose cotransporter 2 inhibitor

Table 1. Selected baseline characteristics after propensity score matching, stratified by baseline cardiovascular disease and diuretic use

Characteristics‡	CVD+/DU+† (n=11,719 pairs)		CVD+/DU- (n=26,092 pairs)		CVD-/DU+ (n=26,864 pairs)		CVD-/DU- (n=155,195 pairs)		Overall (n=219,940 pairs)	
	SGLT2i	DPP4i	SGLT2i	DPP4i	SGLT2i	DPP4i	SGLT2i	DPP4i	SGLT2i	DPP4i
Follow-up, mean (SD) years	1.14 (1.15)	1.27 (1.26)	1.14 (1.14)	1.25 (1.24)	1.30 (1.26)	1.43 (1.33)	1.06 (1.12)	1.11 (1.18)	1.10 (1.14)	1.40 (1.38)
Age, mean (SD), years	64.9 (11.6)	64.7 (12.1)	61.3 (11.1)	61.2 (11.6)	58.4 (12.0)	58.4 (12.3)	51.2 (12.6)	51.1 (12.9)	54.0 (13.1)	53.9 (13.4)
Male, (%)	50.6	51.2	61.3	61.4	50.7	51.2	60.2	60.3	58.6	58.3
Comorbidities, (%)										
Cancer	5.8	5.9	5.1	4.8	5.0	4.4	3.4	3.0	3.9	3.6
Diabetic neuropathy	15.9	16.3	13.4	13.0	11.6	10.6	7.2	6.6	8.9	8.5
Diabetic retinopathy	20.0	19.8	17.9	17.8	14.8	14.0	10.7	9.8	12.6	11.7
Cerebrovascular disease	29.8	29.1	32.0	31.6	-	-	-	-	5.4	4.8
Coronary artery disease	40.2	40.1	42.6	41.8	-	-	-	-	7.2	6.4
Dyslipidemia	40.6	40.6	45.2	45.2	41.3	41.7	39.2	39.1	40.2	40.3
Heart failure	21.3	20.8	5.5	5.2	-	-	-	-	1.8	1.6
Hypertension	75.1	75.3	51.8	51.6	85.0	86.3	32.8	32.1	43.7	43.3
Peripheral vascular disease	30.4	31.5	32.1	33.2	-	-	-	-	5.4	5.0
Ascites	0.2	0.2	0.0	0.0	0.2	0.2	0.0	0.0	0.0	0.0
Cirrhosis	1.1	1.1	0.6	0.6	1.2	1.1	0.5	0.4	0.6	0.6
Acute kidney injury	0.5	0.5	0.2	0.2	0.3	0.3	0.1	0.1	0.2	0.1
Chronic kidney disease	2.6	2.6	1.1	1.0	1.1	1.0	0.4	0.3	0.7	0.5
Chronic respiratory disease	19.3	18.8	13.3	13.2	11.8	11.3	8.1	7.7	9.8	9.3
Use of diuretics, (%)										
Loop diuretics	36.7	36.8	-	-	13.9	13.9	-	-	3.7	3.5
Thiazide diuretics	67.4	68.5	-	-	86.6	87.2	-	-	14.2	13.9
MRA	19.7	17.4	-	-	6.5	5.4	-	-	1.9	1.5
Other diuretics	1.5	1.7	-	-	1.1	1.1	-	-	0.2	0.2
Use of other medications, (%)										
ACEI/ARB	82.8	83.2	51.2	50.4	85.9	86.8	28.2	27.6	40.9	40.2
β-blockers	46.9	46.5	29.6	28.7	21.7	20.7	6.5	5.8	13.3	12.1
Calcium channel blockers	57.9	57.8	45.5	45.0	53.1	53.3	22.7	22.1	31.0	30.4
Nitrates	20.4	20.3	16.7	15.9	1.9	1.7	0.7	0.6	3.8	3.3
Oral antiplatelets	65.0	65.1	61.0	59.8	27.3	26.7	9.7	8.7	20.9	19.3
Oral anticoagulants	9.7	9.4	2.3	2.1	2.4	2.1	0.4	0.3	1.4	1.2
Statins	73.3	72.9	69.6	68.5	58.4	58.3	35.2	34.0	44.1	42.8
Other lipid-lowering agents	26.8	26.6	26.2	25.7	24.3	23.8	15.7	15.2	18.6	18.0
Lv of antidiabetic treatment§, (%)										
1	59.1	59.6	63.5	64.0	60.3	61.6	73.8	75.2	70.1	71.0
2	21.9	21.9	23.3	23.1	27.8	27.5	18.0	1.6	20.0	19.9
3	19.0	18.5	13.2	12.9	11.9	10.8	8.2	0.3	9.9	9.0
Healthcare use										
Hospitalizations										
0	57.0	57.6	66.9	67.6	78.4	79.8	84.2	85.0	80.0	81.0
1-2	36.4	35.9	29.5	29.0	19.1	17.9	14.6	14.1	18.1	17.3
≥3	6.6	6.5	3.6	3.5	2.5	2.3	1.2	0.9	1.9	1.6
Number of outpatient visits										
0-2	0.7	0.7	1.2	1.2	0.7	0.7	9.6	9.8	7.0	7.2
3-5	2.2	2.1	4.0	4.0	3.2	2.9	11.6	11.8	9.2	9.4
≥6	97.1	97.2	94.7	94.8	96.1	96.4	78.8	78.4	83.7	83.4

† Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

‡ Data are number (%) unless otherwise specified. All covariates were assessed during the year before cohort entry, including the cohort entry date.

§ Use of antidiabetic drugs past 365 days before the date of cohort entry: level 1, only one antidiabetic drug or without any antidiabetic drug; level 2, at least two classes of noninsulin antidiabetic drugs; or level 3 at least one insulin treatment as alone or in combination with other antidiabetic drugs.

‖ All of the standardized differences were less than 0.1 after propensity score matching

Abbreviations: ACEI, angiotensin converting enzyme inhibitor; ARB, angiotensin II receptor blocker; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; GLP-1, glucagon-like peptide-1; MRA, mineralocorticoid receptor antagonist; NA, not applicable; SD, standard deviation; SGLT2i, sodium-glucose co-transporter 2 inhibitor

Table 2. Propensity score matched incidence rate of amputation, stratified by baseline cardiovascular disease and diuretic use and overall cohort

	CVD+/DU+[†]		**CVD+/DU-**		**CVD-/DU+**		**CVD-/DU-**		**Overall**	
	SGLT2i	**DPP4i**	**SGLT2i**	**DPP4i**	**SGLT2i**	**DPP4i**	**SGLT2i**	**DPP4i**	**SGLT2i**	**DPP4i**
No. of patients	11,719	11,719	26,092	26,092	26,894	26,894	155,195	155,195	219,940	219,940
No. of events (%)	6 (0.05)	18 (0.15)	9 (0.03)	21 (0.08)	11 (0.04)	22 (0.08)	41 (0.03)	88 (0.06)	67 (0.03)	129 (0.06)
Person years	13,312	14,928	29,708	32,744	34,943	38,456	163,950	172,202	241,948	258,071
Rate per 1000 PY	0.45	1.21	0.30	0.64	0.31	0.57	0.25	0.51	0.28	0.50

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease. In each cohort, SGLT2i users were propensity score matched to DPP4i users with 1:1 ratio based on all potential confounders at baseline.

Abbreviations: CI, confidence interval; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; HR, hazard ratio; SGLT2i, sodium-glucose cotransporter 2 inhibitor; PY, person years

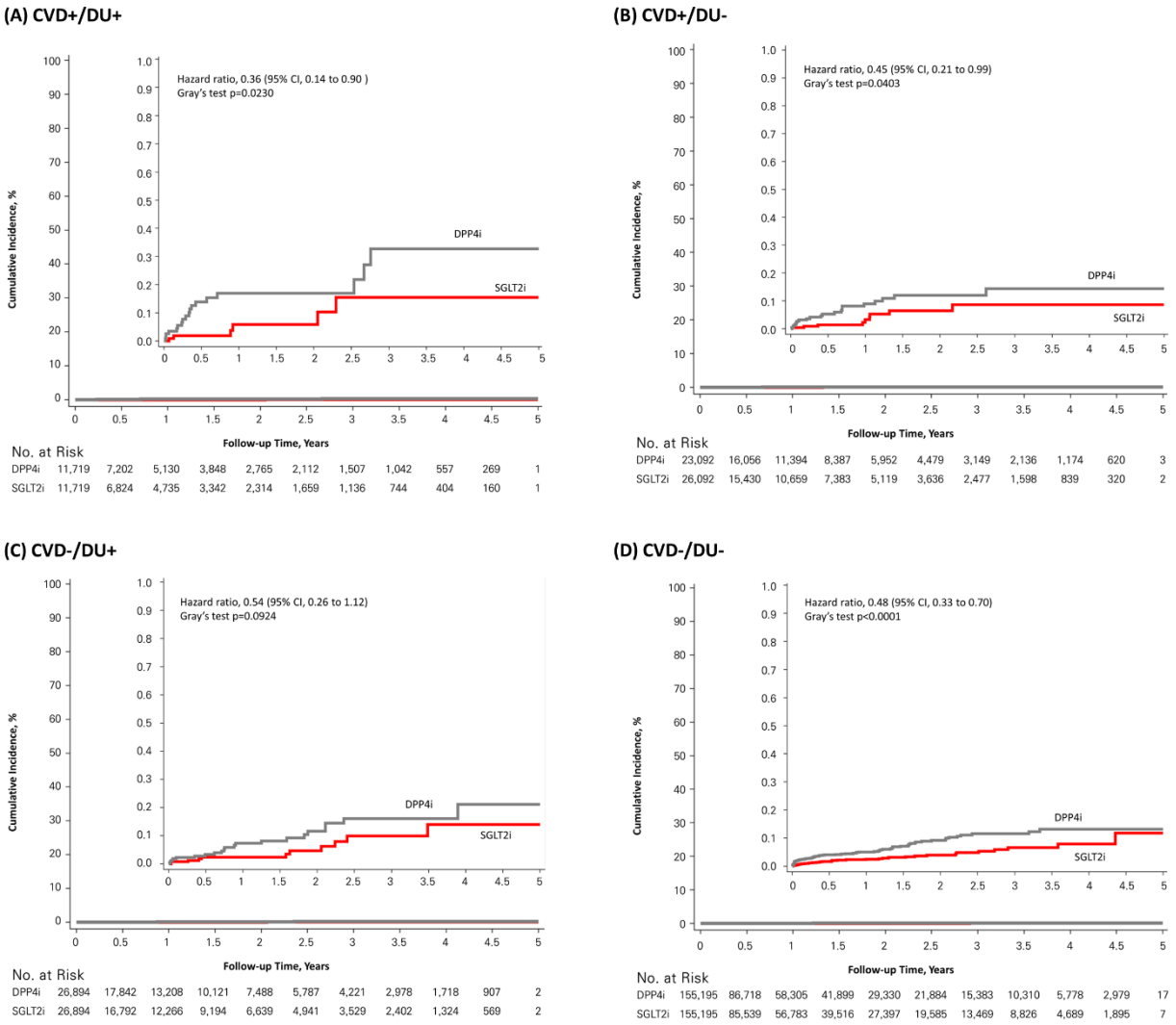


Figure 1. Cumulative incidence curves after propensity score matching for each of the four groups stratified by baseline cardiovascular disease and diuretic use.

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

Abbreviations: CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitors; DU, diuretic; SGLT2i, sodium-glucose cotransporter 2 inhibitor

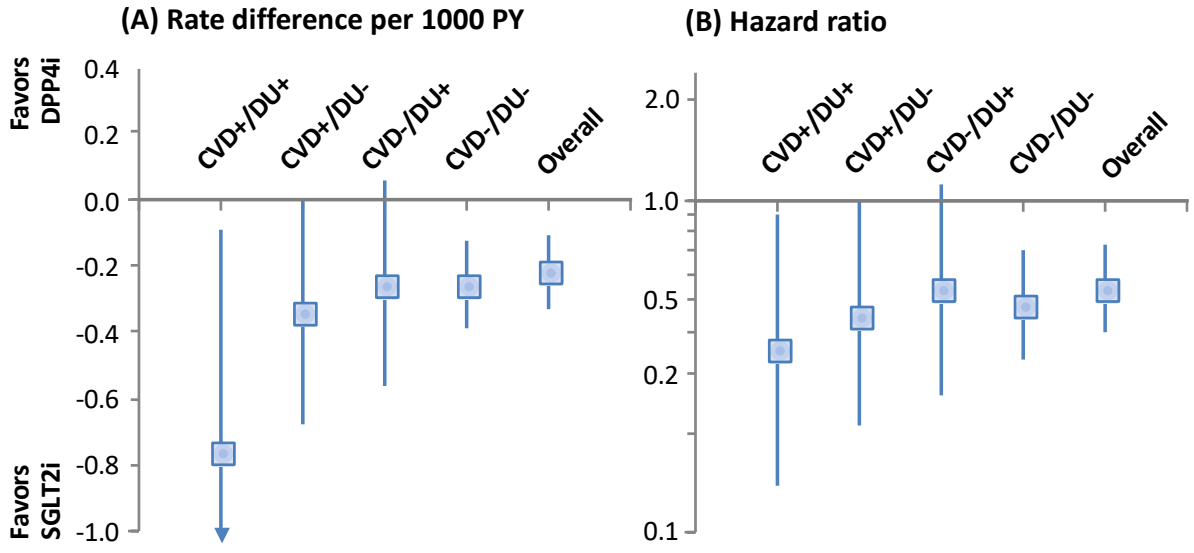


Figure 2. Forest plot for propensity score matched rate differences and hazard ratios of amputation for each of the four groups stratified by baseline cardiovascular disease and diuretic use and overall cohort.

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

Abbreviations: CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; PY, person years; SGLT2i, sodium-glucose cotransporter 2 inhibitor

Supplement

Table S1. ICD-10, ATC, or procedure codes used in study cohort and outcome definition

Table S2. Covariates for generating propensity score

Table S3. Baseline characteristics of SGLT2i and DPP4i users in patients with cardiovascular disease and history of diuretic therapy

Table S4. Baseline characteristics of SGLT2i and DPP4i users in patients with cardiovascular disease and without history of diuretic therapy

Table S5. Baseline characteristics of SGLT2i and DPP4i users in patients without cardiovascular disease and with history of diuretic therapy

Table S6. Baseline characteristics of SGLT2i and DPP4i users in patients without cardiovascular disease and history of diuretic therapy

Table S7. Baseline characteristics of SGLT2i and DPP4i users in overall study cohort

Table S8. Results of stratified and subgroup analyses of rate differences and hazard ratios for amputation associated with the use of SGLT2i compared with DPP4i

Table S9. Results of sensitivity analyses of rate differences and adjusted hazard ratios for amputation associated with the use of SGLT2i compared with DPP4i

Table S10. Results of the hospitalization for heart failure among propensity score matched population

Figure S1. Graphical representation of study design

Figure S2. Flow chart describing selection of the study cohort.

Figure S3. Cumulative incidence curves after propensity score matching for overall cohort.

Table S1. ICD-10, ATC, or procedure codes used in study cohort and outcome definition

Category	**Codes (ICD-10, ATC, or domestic procedure code)**
Stratification of study cohort	
Cardiovascular disease	Cerebrovascular disease: I60.x, I61.x, I62.x, I63.x, I64.x, I65.x, I66.x, I67.x, I68.x, I69.x, G45.0, G45.1, G45.2, G45.3, G45.8, G45.9, G46.x Coronary artery disease: I20.x, I21.x, I22.x Heart failure: I50.x Peripheral vascular disease: I70.x, I73.x
Diuretic use	C03
Outcome	
Amputation	N0571 Amputation for extremity (pelvis) N0572 Amputation for extremity (upper leg) N0573 Amputation for extremity (upper arm, forearm, lower leg) N0574 Amputation for extremity (hand, foot) N0575 Amputation for extremity (finger, toe) *Excluding amputation with diagnosis code of S or T

Table S2. Covariates for generating propensity score

Category	Application
Age	Continuous
Sex	Male, female
Cohort entry year	2016, 2017, 2018, 2019, 2020
Comorbidities	
Cancer	C00.x - C14.x, C15.x - C26.x, C30.x - C39.x, C45.x - C49.x, C50.x, C51.x - C58.x, C60.x - C63.x, C64.x - C68.x, C69.x - C72.x, C73.x - C75.x, C76.x - C80.x, C81.x - C96.x, D00.x - D09.x, D10.x - D36.x, D37.x - D48.x
Diabetic neuropathy	E10.4, E11.4, E13.4, G53.x, G56.x - G59.x, G64.x, G62.9, G90.0, G90.8, G99.0
Diabetic retinopathy	E10.3, E11.3, E13.3, H33.x - H36.x, H54
Cerebrovascular disease	I60.x, I61.x, I62.x, I63.x, I64.x, I65.x, I66.x, I67.x, I68.x, I69.x, G45.0, G45.1, G45.2, G45.3, G45.8, G45.9, G46.x
Coronary artery disease	I20.x, I21.x, I22.x
Dyslipidemia	E78.x
Heart failure	I50.x
Hypertension	I10.x – I15.x
Peripheral vascular disease	I70.x, I73.x
Ascites	R18.x
Cirrhosis	K70.3, K74.x
Acute kidney injury	N17.x
Chronic kidney disease	N18.x, N19.x
Chronic respiratory disease	J84.x, J41.x-J47.x
Use of diuretics	
Loop diuretics	C03CA, C03CB
Thiazide diuretics	C03A
MRA	C03DA
Other diuretics	C03DB
Use of other medications	
ACE inhibitors/ARB	C09A, C09B, C09C, C09D
β-blockers	C07
Calcium channel blockers	C08, C07FB, C09BB, C09DB
Nitrates	C01DA
Oral antiplatelets	B01AC
Oral anticoagulants	B01AF
Statins	C01AA
Other lipid-lowering agents	C01AB, C01AC, C01AD, C01AX
Use of antidiabetic drugs	
Metformin	A10BA02
α-glucosidase inhibitors	A10BF
GLP-1 receptor agonists	A10BJ
Insulin	A10A
Meglitinides	A10BX02, A10BX03, A10BX08
Sulfonylurea	A10BB
Thiazolidinediones	A10BG
Level of antidiabetic treatment	1, 2, 3 *Use of antidiabetic drugs past 365 days before the date of cohort entry: level 1, only one antidiabetic drug or without any antidiabetic drug; level 2, at least two classes of noninsulin antidiabetic drugs; or level 3 at least one insulin treatment as alone or in combination with other antidiabetic drugs
Number of hospitalizations	0, 1-2, ≥3
Number of outpatient visits	0-2, 3-5, ≥6

Table S3. Baseline characteristics of SGLT2i and DPP4i users in patients with cardiovascular disease and diuretic use[†]

Characteristics[‡]	Before matching					After matching				
	SGLT2i		DPP4i		StdDiff[§]	SGLT2i		DPP4i		StdDiff[§]
Number of patients	11,769		93,594			11,719		11,719		
Age, years										
Mean (SD)	64.8	(11.8)	71.3	(10.7)	-0.581	64.9	(11.6)	64.7	(12.1)	0.014
Median [Q1-Q3]	65	[57-74]	73	[64-79]		65	[57-74]	65	[57-74]	
Sex										
Male	5,971	(50.7)	43,047	(46.0)	-0.095	5,935	(50.6)	5,995	(51.2)	0.010
Female	5,798	(49.3)	50,547	(54.0)		5,784	(49.4)	5,724	(48.8)	
SGLT2i molecule										
Dapagliflozin	6,110	(51.92)	-			6,085	(51.9)	-		
Empagliflozin	4,869	(41.37)	-			4,845	(41.3)	-		
Ipragliflozin	121	(1.03)	-			121	(1.0)	-		
Year of cohort entry date										
2016	2,008	(17.1)	27,144	(29.0)	0.381	2,008	(17.1)	2,061	(17.6)	0.028
2017	2,219	(18.9)	20,978	(22.4)		2,219	(18.9)	2,287	(19.5)	
2018	2,149	(18.3)	17,692	(18.9)		2,147	(18.3)	2,080	(17.7)	
2019	2,701	(23.0)	15,206	(16.2)		2,676	(22.8)	2,712	(23.1)	
2020	2,692	(22.9)	12,574	(13.4)		2,669	(22.8)	2,579	(22.0)	
Comorbidities										
Cancer	684	(5.8)	8,037	(8.6)	-0.108	684	(5.8)	690	(5.9)	-0.002
Diabetic nephropathy	1,873	(15.9)	16,909	(18.1)	-0.057	1,869	(15.9)	1,909	(16.3)	-0.009
Diabetic neuropathy	2,351	(20.0)	20,215	(21.6)	-0.040	2,348	(20.0)	2,317	(19.8)	0.007
Diabetic retinopathy	3,489	(29.6)	37,335	(39.9)	-0.216	3,488	(29.8)	3,414	(29.1)	0.014
Cerebrovascular disease	4,739	(40.3)	31,659	(33.8)	0.134	4,716	(40.2)	4,702	(40.1)	0.002
Coronary artery disease	4,776	(40.6)	34,986	(37.4)	0.066	4,755	(40.6)	4,761	(40.6)	-0.001
Dyslipidemia	2,521	(21.4)	17,972	(19.2)	0.055	2,492	(21.3)	2,436	(20.8)	0.012
Heart failure	8,833	(75.1)	72,697	(77.7)	-0.062	8,802	(75.1)	8,828	(75.3)	-0.005
Hypertension	3,567	(30.3)	28,385	(30.3)	0.000	3,560	(30.4)	3,690	(31.5)	-0.024
Peripheral vascular disease	22	(0.2)	253	(0.3)	-0.017	21	(0.2)	24	(0.2)	-0.006
Cirrhosis	134	(1.1)	1,557	(1.7)	-0.045	133	(1.1)	125	(1.1)	0.007
Acute kidney injury	56	(0.5)	1,243	(1.3)	-0.090	56	(0.5)	62	(0.5)	-0.007
Chronic kidney disease	301	(2.6)	5,120	(5.5)	-0.149	301	(2.6)	306	(2.6)	-0.003
Chronic respiratory disease	2,268	(19.3)	20,407	(21.8)		2,256	(19.3)	2,207	(18.8)	0.011
Use of diuretics										
Loop diuretics	4,333	(36.8)	37,510	(40.1)	-0.067	4,306	(36.7)	4,308	(36.8)	0.000
Thiazide diuretics	7,916	(67.3)	62,313	(66.6)	0.015	7,898	(67.4)	8,025	(68.5)	-0.023
MRA	2,339	(19.9)	16,003	(17.1)	0.072	2,310	(19.7)	2,042	(17.4)	0.059
Other diuretics	177	(1.5)	1,601	(1.7)	-0.016	177	(1.5)	200	(1.7)	-0.016
Use of other medications										
ACEI/ARB	9,757	(82.9)	73,293	(78.3)	0.116	9,708	(82.8)	9,747	(83.2)	-0.009
β-blockers	5,542	(47.1)	38,445	(41.1)	0.121	5,502	(46.9)	5,450	(46.5)	0.009
Calcium channel blockers	6,822	(58.0)	55,532	(59.3)	-0.028	6,791	(57.9)	6,771	(57.8)	0.003
Nitrates	2,410	(20.5)	15,262	(16.3)	0.108	2,393	(20.4)	2,384	(20.3)	0.002
Oral antiplatelets	7,646	(65.0)	63,401	(67.7)	-0.059	7,621	(65.0)	7,627	(65.1)	-0.001
Oral anticoagulants	1,139	(9.7)	8,526	(9.1)	0.019	1,135	(9.7)	1,097	(9.4)	0.011
Statins	8,626	(73.3)	62,651	(66.9)	0.139	8,585	(73.3)	8,542	(72.9)	0.008
Other lipid-lowering agents	3,159	(26.8)	18,867	(20.2)	0.158	3,136	(26.8)	3,116	(26.6)	0.004
Use of antidiabetic drugs										
Metformin	6,971	(59.2)	56,653	(60.5)	-0.026	6,950	(59.3)	6,908	(58.9)	0.007
α-glucosidase inhibitors	416	(3.5)	5,265	(5.6)	-0.100	416	(3.5)	382	(3.3)	0.016
GLP-1 receptor agonists	59	(0.5)	119	(0.1)	0.067	56	(0.5)	49	(0.4)	0.009

Insulin	2,241	(19.0)	21,434	(22.9)	-0.095	2,227	(19.0)	2,170	(18.5)	0.012
Meglitinides	87	(0.7)	1,242	(1.3)	-0.058	87	(0.7)	92	(0.8)	-0.005
Sulfonylurea	3,505	(29.8)	35,195	(37.6)	-0.166	3,503	(29.9)	3,453	(29.5)	0.009
Thiazolidinediones	1,041	(8.8)	6,569	(7.0)	0.068	1,036	(8.8)	1,029	(8.8)	0.002
Lv of antidiabetic treatment‖										
1	6,952	(59.1)	48,792	(52.1)	0.141	6,921	(59.1)	6,987	(59.6)	0.014
2	2,576	(21.9)	23,368	(25.0)		2,571	(21.9)	2,562	(21.9)	
3	2,241	(19.0)	21,434	(22.9)		2,227	(19.0)	2,170	(18.5)	
Healthcare use										
Hospitalizations										
0	6,695	(56.9)	48,465	(51.8)	0.133	6,676	(57.0)	6,749	(57.6)	0.013
1-2	4,290	(36.5)	35,974	(38.4)		4,264	(36.4)	4,205	(35.9)	
≥3	784	(6.7)	9,155	(9.8)		779	(6.6)	765	(6.5)	
Number of outpatient visits										
0-2	81	(0.7)	596	(0.6)	0.062	81	(0.7)	85	(0.7)	0.008
3-5	254	(2.2)	1,263	(1.3)		253	(2.2)	242	(2.1)	
≥6	11,434	(97.2)	91,735	(98.0)		11,385	(97.1)	11,392	(97.2)	

† Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

‡ Data are number (%) unless otherwise specified. All covariates were assessed during the year before cohort entry, including the cohort entry date.

§ Red-colored number indicates negligible difference (absolute standardized mean difference <0.1)

‖ Use of antidiabetic drugs past 365 days before the date of cohort entry: level 1, only one antidiabetic drug or without any antidiabetic drug; level 2, at least two classes of noninsulin antidiabetic drugs; or level 3 at least one insulin treatment as alone or in combination with other antidiabetic drugs.

Abbreviations: ACEI, angiotensin converting enzyme inhibitor; ARB, angiotensin II receptor blocker; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; GLP-1, glucagon-like peptide-1; MRA, mineralocorticoid receptor antagonist; NA, not applicable; SD, standard deviation; SGLT2i, sodium-glucose co-transporter 2 inhibitor

Table S4. Baseline characteristics of SGLT2i and DPP4i users in patients with cardiovascular disease only (no diuretic use)†

Characteristics‡	Before matching					After matching				
	SGLT2i		DPP4i		StdDiff§	SGLT2i		DPP4i		StdDiff§
Number of patients	26,096		182,673			26,092		26,092		
Age, years										
Mean (SD)	61.3	(11.1)	67.6	(11.0)	-0.562	61.3	(11.1)	61.2	(11.6)	0.013
Median [Q1-Q3]	62	[54-69]	68	[60-76]		62	[54-69]	61	[54-69]	
Sex										
Male	15,992	(61.3)	102,508	(56.1)	-0.105	15,988	(61.3)	16,028	(61.4)	0.003
Female	10,104	(38.7)	80,165	(43.9)		10,104	(38.7)	10,064	(38.6)	
SGLT2i molecule										
Dapagliflozin	13,848	(53.07)	-			13,848	(53.1)	-		
Empagliflozin	10,428	(39.96)	-			10,424	(40.0)	-		
Ipragliflozin	328	(1.26)	-			328	(1.3)	-		
Year of cohort entry date										
2016	4,270	(16.4)	49,614	(27.2)	0.354	4,270	(16.4)	4,206	(16.1)	0.019
2017	4,837	(18.5)	40,011	(21.9)		4,837	(18.5)	4,839	(18.5)	
2018	4,576	(17.5)	34,442	(18.9)		4,576	(17.5)	4,600	(17.6)	
2019	6,338	(24.3)	31,542	(17.3)		6,334	(24.3)	6,514	(25.0)	
2020	6,075	(23.3)	27,064	(14.8)		6,075	(23.3)	5,933	(22.7)	
Comorbidities										
Cancer	1,326	(5.1)	13,084	(7.2)	-0.087	1,326	(5.1)	1,246	(4.8)	0.014
Diabetic nephropathy	3,488	(13.4)	29,088	(15.9)	-0.072	3,488	(13.4)	3,386	(13.0)	0.012
Diabetic neuropathy	4,679	(17.9)	36,722	(20.1)	-0.055	4,678	(17.9)	4,646	(17.8)	0.003
Diabetic retinopathy	8,361	(32.0)	78,501	(43.0)	-0.227	8,361	(32.0)	8,250	(31.6)	0.009
Cerebrovascular disease	11,118	(42.6)	57,424	(31.4)	0.233	11,115	(42.6)	10,901	(41.8)	0.017
Coronary artery disease	11,802	(45.2)	75,638	(41.4)	0.077	11,799	(45.2)	11,798	(45.2)	0.000
Dyslipidemia	1,448	(5.5)	8,054	(4.4)	0.052	1,447	(5.5)	1,366	(5.2)	0.014
Heart failure	13,529	(51.8)	98,726	(54.0)	-0.044	13,526	(51.8)	13,468	(51.6)	0.004
Hypertension	8,371	(32.1)	63,168	(34.6)	-0.053	8,371	(32.1)	8,672	(33.2)	-0.025
Peripheral vascular disease	2	(0.0)	33	(0.0)	-0.009	2	(0.0)	2	(0.0)	0.000
Cirrhosis	161	(0.6)	1,613	(0.9)	-0.031	161	(0.6)	153	(0.6)	0.004
Acute kidney injury	57	(0.2)	711	(0.4)	-0.031	57	(0.2)	60	(0.2)	-0.002
Chronic kidney disease	280	(1.1)	3,976	(2.2)	-0.087	280	(1.1)	265	(1.0)	0.006
Chronic respiratory disease	3,482	(13.3)	28,776	(15.8)	-0.068	3,481	(13.3)	3,434	(13.2)	0.005
Use of diuretics										
Loop diuretics	-		-			-		-		
Thiazide diuretics	-		-			-		-		
MRA	-		-			-		-		
Other diuretics	-		-			-		-		
Use of other medications										
ACEI/ARB	13,370	(51.2)	84,456	(46.2)	0.100	13,367	(51.2)	13,163	(50.4)	0.016
β-blockers	7,738	(29.7)	43,415	(23.8)	0.133	7,736	(29.6)	7,476	(28.7)	0.022
Calcium channel blockers	11,883	(45.5)	84,084	(46.0)	-0.010	11,880	(45.5)	11,739	(45.0)	0.011
Nitrates	4,358	(16.7)	21,367	(11.7)	0.144	4,356	(16.7)	4,159	(15.9)	0.020
Oral antiplatelets	15,913	(61.0)	112,731	(61.7)	-0.015	15,911	(61.0)	15,612	(59.8)	0.023
Oral anticoagulants	602	(2.3)	4,578	(2.5)	-0.013	602	(2.3)	556	(2.1)	0.012
Statins	18,167	(69.6)	115,377	(63.2)	0.137	18,163	(69.6)	17,871	(68.5)	0.024
Other lipid-lowering agents	6,844	(26.2)	35,098	(19.2)	0.168	6,842	(26.2)	6,714	(25.7)	0.011
Use of antidiabetic drugs										
Metformin	15,946	(61.1)	114,607	(62.7)	-0.034	15,944	(61.1)	15,816	(60.6)	0.010
α-glucosidase inhibitors	904	(3.5)	9,537	(5.2)	-0.086	904	(3.5)	834	(3.2)	0.015
GLP-1 receptor agonists	117	(0.4)	239	(0.1)	0.059	116	(0.4)	110	(0.4)	0.004

Insulin	3,441	(13.2)	25,870	(14.2)	-0.028	3,441	(13.2)	3,367	(12.9)	0.008
Meglitinides	164	(0.6)	1,826	(1.0)	-0.041	164	(0.6)	163	(0.6)	0.000
Sulfonylurea	7,208	(27.6)	63,150	(34.6)	-0.151	7,208	(27.6)	7,181	(27.5)	0.002
Thiazolidinediones	2,141	(8.2)	11,715	(6.4)	0.069	2,140	(8.2)	2,120	(8.1)	0.003
Lv of antidiabetic treatment‖										
1	16,580	(63.5)	107,217	(58.7)	0.102	16,577	(63.5)	16,709	(64.0)	0.011
2	6,075	(23.3)	49,586	(27.1)		6,074	(23.3)	6,016	(23.1)	
3	3,441	(13.2)	25,870	(14.2)		3,441	(13.2)	3,367	(12.9)	
Healthcare use										
Hospitalizations										
0	17,464	(66.9)	116,713	(63.9)	0.076	17,462	(66.9)	17,628	(67.6)	0.014
1-2	7,694	(29.5)	57,278	(31.4)		7,692	(29.5)	7,556	(29.0)	
≥3	938	(3.6)	8,682	(4.8)		938	(3.6)	908	(3.5)	
Number of outpatient visits										
0-2	325	(1.2)	2,811	(1.5)	0.072	325	(1.2)	316	(1.2)	0.005
3-5	1,055	(4.0)	5,142	(2.8)		1,055	(4.0)	1,037	(4.0)	
≥6	24,716	(94.7)	174,720	(95.6)		24,712	(94.7)	24,739	(94.8)	

† Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

‡ Data are number (%) unless otherwise specified. All covariates were assessed during the year before cohort entry, including the cohort entry date.

§ Red-colored number indicates negligible difference (absolute standardized mean difference <0.1)

‖ Use of antidiabetic drugs past 365 days before the date of cohort entry: level 1, only one antidiabetic drug or without any antidiabetic drug; level 2, at least two classes of noninsulin antidiabetic drugs; or level 3 at least one insulin treatment as alone or in combination with other antidiabetic drugs.

Abbreviations: ACEI, angiotensin converting enzyme inhibitor; ARB, angiotensin II receptor blocker; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; GLP-1, glucagon-like peptide-1; MRA, mineralocorticoid receptor antagonist; NA, not applicable; SD, standard deviation; SGLT2i, sodium-glucose co-transporter 2 inhibitor

Table S5. Baseline characteristics of SGLT2i and DPP4i users in patients with diuretic use only (no cardiovascular disease)[†]

Characteristics[‡]	Before matching					After matching				
	SGLT2i		DPP4i		StdDiff[§]	SGLT2i		DPP4i		StdDiff[§]
Number of patients	26,900		188,446			26,894		26,894		
Age, years										
Mean (SD)	58.4	(12.0)	65.7	(12.1)	-0.605	58.4	(12.0)	58.4	(12.3)	0.002
Median [Q1-Q3]	59	[50-66]	66	[57-75]		59	[50-66]	58	[50-67]	
Sex										
Male	13,627	(50.7)	93,189	(49.5)	-0.024	13,624	(50.7)	13,772	(51.2)	0.011
Female	13,273	(49.3)	95,257	(50.5)		13,270	(49.3)	13,122	(48.8)	
SGLT2i molecule										
Dapagliflozin	14,511	(53.94)	-			14,507	(53.9)	-		
Empagliflozin	10,525	(39.13)	-			10,524	(39.1)	-		
Ipragliflozin	363	(1.35)	-			363	(1.3)	-		
Year of cohort entry date										
2016	5,208	(19.4)	54,729	(29.0)	0.296	5,207	(19.4)	5,078	(18.9)	0.015
2017	5,524	(20.5)	42,685	(22.7)		5,523	(20.5)	5,550	(20.6)	
2018	4,844	(18.0)	35,247	(18.7)		4,844	(18.0)	4,864	(18.1)	
2019	5,847	(21.7)	30,342	(16.1)		5,846	(21.7)	5,960	(22.2)	
2020	5,477	(20.4)	25,443	(13.5)		5,474	(20.4)	5,442	(20.2)	
Comorbidities										
Cancer	1,336	(5.0)	16,088	(8.5)	-0.143	1,335	(5.0)	1,186	(4.4)	0.026
Diabetic nephropathy	3,114	(11.6)	25,670	(13.6)	-0.062	3,114	(11.6)	2,856	(10.6)	0.031
Diabetic neuropathy	3,991	(14.8)	31,413	(16.7)	-0.050	3,989	(14.8)	3,756	(14.0)	0.025
Diabetic retinopathy	-		-			-		-		
Cerebrovascular disease	-		-			-		-		
Coronary artery disease	11,109	(41.3)	64,772	(34.4)	0.143	11,105	(41.3)	11,211	(41.7)	-0.008
Dyslipidemia	-		-			-		-		
Heart failure	22,851	(84.9)	159,174	(84.5)	0.013	22,849	(85.0)	23,208	(86.3)	-0.038
Hypertension	-		-			-		-		
Peripheral vascular disease	44	(0.2)	954	(0.5)	-0.059	44	(0.2)	41	(0.2)	0.003
Cirrhosis	323	(1.2)	5,354	(2.8)	-0.117	323	(1.2)	303	(1.1)	0.007
Acute kidney injury	75	(0.3)	1,297	(0.7)	-0.059	75	(0.3)	70	(0.3)	0.004
Chronic kidney disease	296	(1.1)	5,288	(2.8)	-0.123	296	(1.1)	277	(1.0)	0.007
Chronic respiratory disease	3,162	(11.8)	26,302	(14.0)	-0.066	3,161	(11.8)	3,036	(11.3)	0.015
Use of diuretics										
Loop diuretics	3,740	(13.9)	36,539	(19.4)	-0.148	3,739	(13.9)	3,729	(13.9)	0.001
Thiazide diuretics	23,295	(86.6)	155,758	(82.7)	0.110	23,290	(86.6)	23,465	(87.2)	-0.019
MRA	1,752	(6.5)	13,678	(7.3)	-0.029	1,752	(6.5)	1,441	(5.4)	0.049
Other diuretics	309	(1.1)	2,326	(1.2)	-0.008	308	(1.1)	286	(1.1)	0.008
Use of other medications										
ACEI/ARB	23,115	(85.9)	152,076	(80.7)	0.141	23,110	(85.9)	23,353	(86.8)	-0.026
β-blockers	5,829	(21.7)	38,619	(20.5)	0.029	5,828	(21.7)	5,576	(20.7)	0.023
Calcium channel blockers	14,289	(53.1)	99,932	(53.0)	0.002	14,286	(53.1)	14,340	(53.3)	-0.004
Nitrates	502	(1.9)	3,663	(1.9)	-0.006	502	(1.9)	449	(1.7)	0.015
Oral antiplatelets	7,349	(27.3)	59,293	(31.5)	-0.091	7,348	(27.3)	7,192	(26.7)	0.013
Oral anticoagulants	643	(2.4)	4,818	(2.6)	-0.011	643	(2.4)	576	(2.1)	0.017
Statins	15,714	(58.4)	97,497	(51.7)	0.135	15,710	(58.4)	15,678	(58.3)	0.002
Other lipid-lowering agents	6,537	(24.3)	33,431	(17.7)	0.162	6,534	(24.3)	6,412	(23.8)	0.011
Use of antidiabetic drugs										
Metformin	16,540	(61.5)	121,885	(64.7)	-0.066	16,534	(61.5)	16,379	(60.9)	0.012
α-glucosidase inhibitors	934	(3.5)	9,485	(5.0)	-0.077	933	(3.5)	820	(3.0)	0.024
GLP-1 receptor agonists	132	(0.5)	239	(0.1)	0.066	126	(0.5)	105	(0.4)	0.012

Insulin	3,214	(11.9)	27,689	(14.7)	-0.081	3,211	(11.9)	2,913	(10.8)	0.035
Meglitinides	133	(0.5)	1,358	(0.7)	-0.029	132	(0.5)	121	(0.4)	0.006
Sulfonylurea	8,577	(31.9)	74,857	(39.7)	-0.164	8,574	(31.9)	8,430	(31.3)	0.012
Thiazolidinediones	2,686	(10.0)	13,710	(7.3)	0.097	2,684	(10.0)	2,455	(9.1)	0.029
Lv of antidiabetic treatment‖										
1	16,210	(60.3)	101,595	(53.9)	0.131	16,210	(60.3)	16,577	(61.6)	0.038
2	7,476	(27.8)	59,162	(31.4)		7,473	(27.8)	7,404	(27.5)	
3	3,214	(11.9)	27,689	(14.7)		3,211	(11.9)	2,913	(10.8)	
Healthcare use										
Hospitalizations										
0	21,090	(78.4)	134,056	(71.1)	0.194	21,086	(78.4)	21,453	(79.8)	0.034
1-2	5,134	(19.1)	44,211	(23.5)		5,132	(19.1)	4,824	(17.9)	
≥3	676	(2.5)	10,179	(5.4)		676	(2.5)	617	(2.3)	
Number of outpatient visits										
0-2	198	(0.7)	1,570	(0.8)	0.044	198	(0.7)	189	(0.7)	0.019
3-5	861	(3.2)	4,703	(2.5)		861	(3.2)	777	(2.9)	
≥6	25,841	(96.1)	182,173	(96.7)		25,835	(96.1)	25,928	(96.4)	

† Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

‡ Data are number (%) unless otherwise specified. All covariates were assessed during the year before cohort entry, including the cohort entry date. /

§ Red-colored number indicates negligible difference (absolute standardized mean difference <0.1)

‖ Use of antidiabetic drugs past 365 days before the date of cohort entry: level 1, only one antidiabetic drug or without any antidiabetic drug; level 2, at least two classes of noninsulin antidiabetic drugs; or level 3 at least one insulin treatment as alone or in combination with other antidiabetic drugs.

Abbreviations: ACEI, angiotensin converting enzyme inhibitor; ARB, angiotensin II receptor blocker; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; GLP-1, glucagon-like peptide-1; MRA, mineralocorticoid receptor antagonist; NA, not applicable; SD, standard deviation; SGLT2i, sodium-glucose co-transporter 2 inhibitor

Table S6. Baseline characteristics of SGLT2i and DPP4i users in patients without cardiovascular disease and diuretic use[†]

Characteristics[‡]	Before matching					After matching				
	SGLT2i		DPP4i		StdDiff[§]	SGLT2i		DPP4i		StdDiff[§]
Number of patients	155,211		904,108			155,195		155,195		
Age, years										
Mean (SD)	51.2	(12.6)	58.0	(13.2)	-0.528	51.2	(12.6)	51.1	(12.9)	0.002
Median [Q1-Q3]	52	[43-60]	58	[49-67]		52	[43-60]	52	[42-60]	
Sex										
Male	93,404	(60.2)	550,812	(60.9)	0.015	93,395	(60.2)	93,538	(60.3)	0.002
Female	61,807	(39.8)	353,296	(39.1)		61,800	(39.8)	61,657	(39.7)	
SGLT2i molecule										
Dapagliflozin	86,154	(55.51)	-			86,147	(55.5)	-		
Empagliflozin	58,578	(37.74)	-			58,570	(37.7)	-		
Ipragliflozin	2,159	(1.39)	-			2,159	(1.4)	-		
Year of cohort entry date										
2016	23,582	(15.2)	238,185	(26.3)	0.363	23,582	(15.2)	22,502	(14.5)	0.022
2017	28,397	(18.3)	196,670	(21.8)		28,397	(18.3)	28,151	(18.1)	
2018	28,357	(18.3)	173,483	(19.2)		28,357	(18.3)	28,319	(18.2)	
2019	37,330	(24.1)	157,105	(17.4)		37,319	(24.0)	37,952	(24.5)	
2020	37,545	(24.2)	138,665	(15.3)		37,540	(24.2)	38,271	(24.7)	
Comorbidities										
Cancer	5,203	(3.4)	45,556	(5.0)	-0.084	5,203	(3.4)	4,623	(3.0)	0.021
Diabetic nephropathy	11,139	(7.2)	79,341	(8.8)	-0.059	11,136	(7.2)	10,225	(6.6)	0.023
Diabetic neuropathy	16,680	(10.7)	106,891	(11.8)	-0.034	16,675	(10.7)	15,251	(9.8)	0.030
Diabetic retinopathy	-		-			-		-		
Cerebrovascular disease	-		-			-		-		
Coronary artery disease	60,814	(39.2)	314,954	(34.8)	0.090	60,805	(39.2)	60,652	(39.1)	0.002
Dyslipidemia	-		-			-		-		
Heart failure	50,956	(32.8)	312,478	(34.6)	-0.037	50,953	(32.8)	49,860	(32.1)	0.015
Hypertension	-		-			-		-		
Peripheral vascular disease	22	(0.0)	230	(0.0)	-0.008	22	(0.0)	17	(0.0)	0.003
Cirrhosis	750	(0.5)	8,250	(0.9)	-0.052	750	(0.5)	635	(0.4)	0.011
Acute kidney injury	162	(0.1)	1,514	(0.2)	-0.017	162	(0.1)	145	(0.1)	0.003
Chronic kidney disease	566	(0.4)	6,886	(0.8)	-0.053	566	(0.4)	450	(0.3)	0.013
Chronic respiratory disease	12,543	(8.1)	83,364	(9.2)	-0.041	12,543	(8.1)	11,962	(7.7)	0.014
Use of diuretics										
Loop diuretics	-		-			-		-		
Thiazide diuretics	-		-			-		-		
MRA	-		-			-		-		
Other diuretics	-		-			-		-		
Use of other medications										
ACEI/ARB	43,724	(28.2)	239,971	(26.5)	0.037	43,715	(28.2)	42,768	(27.6)	0.014
β-blockers	10,124	(6.5)	59,121	(6.5)	-0.001	10,123	(6.5)	8,931	(5.8)	0.032
Calcium channel blockers	35,285	(22.7)	217,511	(24.1)	-0.031	35,282	(22.7)	34,358	(22.1)	0.014
Nitrates	1,052	(0.7)	5,548	(0.6)	0.008	1,052	(0.7)	875	(0.6)	0.015
Oral antiplatelets	14,984	(9.7)	109,348	(12.1)	-0.078	14,984	(9.7)	13,522	(8.7)	0.033
Oral anticoagulants	624	(0.4)	4,002	(0.4)	-0.006	624	(0.4)	516	(0.3)	0.012
Statins	54,582	(35.2)	299,420	(33.1)	0.043	54,573	(35.2)	52,784	(34.0)	0.024
Other lipid-lowering agents	24,445	(15.7)	110,721	(12.2)	0.101	24,440	(15.7)	23,641	(15.2)	0.014
Use of antidiabetic drugs										
Metformin	74,473	(48.0)	464,269	(51.4)	-0.067	74,460	(48.0)	72,939	(47.0)	0.020
α-glucosidase inhibitors	2,726	(1.8)	25,514	(2.8)	-0.071	2,726	(1.8)	2,428	(1.6)	0.015
GLP-1 receptor agonists	699	(0.5)	933	(0.1)	0.066	687	(0.4)	512	(0.3)	0.018

Insulin	12,812	(8.3)	68,493	(7.6)	0.025	12,798	(8.2)	11,128	(7.2)	0.040
Meglitinides	473	(0.3)	4,068	(0.4)	-0.024	473	(0.3)	383	(0.2)	0.011
Sulfonylurea	30,583	(19.7)	233,010	(25.8)	-0.145	30,582	(19.7)	29,690	(19.1)	0.015
Thiazolidinediones	8,827	(5.7)	41,401	(4.6)	0.050	8,822	(5.7)	7,923	(5.1)	0.026
Lv of antidiabetic treatment‖										
1	114,502	(73.8)	633,545	(70.1)	0.110	114,502	(73.8)	116,687	(75.2)	0.043
2	27,897	(18.0)	202,070	(22.4)		27,895	(18.0)	27,380	(17.6)	
3	12,812	(8.3)	68,493	(7.6)		12,798	(8.2)	11,128	(7.2)	
Healthcare use										
Hospitalizations										
0	130,716	(84.2)	741,747	(82.0)	0.072	130,703	(84.2)	131,896	(85.0)	0.028
1-2	22,696	(14.6)	145,638	(16.1)		22,694	(14.6)	21,867	(14.1)	
≥3	1,799	(1.2)	16,723	(1.8)		1,798	(1.2)	1,432	(0.9)	
Number of outpatient visits										
0-2	14,901	(9.6)	92,542	(10.2)	0.045	14,901	(9.6)	15,266	(9.8)	0.010
3-5	18,078	(11.6)	93,571	(10.3)		18,077	(11.6)	18,311	(11.8)	
≥6	122,232	(78.8)	717,995	(79.4)		122,217	(78.8)	121,618	(78.4)	

† Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

‡ Data are number (%) unless otherwise specified. All covariates were assessed during the year before cohort entry, including the cohort entry date.

§ Red-colored number indicates negligible difference (absolute standardized mean difference <0.1)

‖ Use of antidiabetic drugs past 365 days before the date of cohort entry: level 1, only one antidiabetic drug or without any antidiabetic drug; level 2, at least two classes of noninsulin antidiabetic drugs; or level 3 at least one insulin treatment as alone or in combination with other antidiabetic drugs.

Abbreviations: ACEI, angiotensin converting enzyme inhibitor; ARB, angiotensin II receptor blocker; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; GLP-1, glucagon-like peptide-1; MRA, mineralocorticoid receptor antagonist; SD, standard deviation; SGLT2i, sodium-glucose co-transporter 2 inhibitor

Table S7. Baseline characteristics of SGLT2i and DPP4i users in overall study cohort

Characteristics†	Before matching					After matching				
	SGLT2i		DPP4i		StdDiff‡	SGLT2i		DPP4i		StdDiff‡
Number of patients	219,976		1,368,821			219,940		219,940		
Age, years										
Mean (SD)	54.0	(13.1)	61.2	(13.5)	-0.545	54.0	(13.1)	53.9	(13.4)	0.008
Median [Q1-Q3]	54	[45-63]	61	[52-71]		54	[45-63]	54	[45-63]	
Sex										
Male	128,994	(58.6)	789,556	(57.7)	-0.019	128,976	(58.6)	128,330	(58.3)	-0.006
Female	90,982	(41.4)	579,265	(42.3)		90,964	(41.4)	91,610	(41.7)	
SGLT2i molecule										
Dapagliflozin	120,623	(54.83)	-			120,603	(54.8)	-		
Empagliflozin	84,400	(38.37)	-			84,385	(38.4)	-		
Ipragliflozin	2,971	(1.35)	-			2,971	(1.4)	-		
Year of cohort entry date										
2016	35,068	(15.9)	369,672	(27.0)	0.357	35,067	(15.9)	33,838	(15.4)	0.016
2017	40,977	(18.6)	300,344	(21.9)		40,977	(18.6)	41,081	(18.7)	
2018	39,926	(18.2)	260,864	(19.1)		39,923	(18.2)	40,073	(18.2)	
2019	52,216	(23.7)	234,195	(17.1)		52,205	(23.7)	52,689	(24.0)	
2020	51,789	(23.5)	203,746	(14.9)		51,768	(23.5)	52,259	(23.8)	
Comorbidities										
Cancer	8,549	(3.9)	82,765	(6.0)	-0.100	8,549	(3.9)	7,971	(3.6)	0.014
Diabetic nephropathy	19,614	(8.9)	151,008	(11.0)	-0.071	19,613	(8.9)	18,585	(8.5)	0.017
Diabetic neuropathy	27,701	(12.6)	195,241	(14.3)	-0.049	27,690	(12.6)	25,762	(11.7)	0.027
Diabetic retinopathy	11,850	(5.4)	115,836	(8.5)	-0.121	11,849	(5.4)	10,619	(4.8)	0.025
Cerebrovascular disease	15,857	(7.2)	89,083	(6.5)	0.028	15,856	(7.2)	14,084	(6.4)	0.032
Coronary artery disease	88,501	(40.2)	490,350	(35.8)	0.091	88,486	(40.2)	88,636	(40.3)	-0.001
Dyslipidemia	3,969	(1.8)	26,026	(1.9)	-0.007	3,969	(1.8)	3,415	(1.6)	0.020
Heart failure	96,169	(43.7)	643,075	(47.0)	-0.066	96,159	(43.7)	95,249	(43.3)	0.008
Hypertension	11,938	(5.4)	91,553	(6.7)	-0.053	11,938	(5.4)	11,062	(5.0)	0.018
Peripheral vascular disease	90	(0.0)	1,470	(0.1)	-0.024	90	(0.0)	82	(0.0)	0.002
Cirrhosis	1,368	(0.6)	16,774	(1.2)	-0.063	1,368	(0.6)	1,238	(0.6)	0.008
Acute kidney injury	350	(0.2)	4,765	(0.3)	-0.038	350	(0.2)	283	(0.1)	0.008
Chronic kidney disease	1,443	(0.7)	21,270	(1.6)	-0.086	1,443	(0.7)	1,194	(0.5)	0.015
Chronic respiratory disease	21,455	(9.8)	158,849	(11.6)	-0.060	21,453	(9.8)	20,465	(9.3)	0.015
Use of diuretics										
Loop diuretics	8,073	(3.7)	74,049	(5.4)	-0.084	8,071	(3.7)	7,621	(3.5)	0.011
Thiazide diuretics	31,211	(14.2)	218,071	(15.9)	-0.049	31,207	(14.2)	30,546	(13.9)	0.009
MRA	4,091	(1.9)	29,681	(2.2)	-0.022	4,091	(1.9)	3,279	(1.5)	0.029
Other diuretics	486	(0.2)	3,927	(0.3)	-0.013	486	(0.2)	443	(0.2)	0.004
Use of other medications										
ACEI/ARB	89,966	(40.9)	549,796	(40.2)	0.015	89,952	(40.9)	88,458	(40.2)	0.014
β-blockers	29,233	(13.3)	179,600	(13.1)	0.005	29,229	(13.3)	26,683	(12.1)	0.035
Calcium channel blockers	68,279	(31.0)	457,059	(33.4)	-0.050	68,266	(31.0)	66,945	(30.4)	0.013
Nitrates	8,322	(3.8)	45,840	(3.3)	0.023	8,321	(3.8)	7,333	(3.3)	0.024
Oral antiplatelets	45,892	(20.9)	344,773	(25.2)	-0.103	45,890	(20.9)	42,377	(19.3)	0.040
Oral anticoagulants	3,008	(1.4)	21,924	(1.6)	-0.019	3,008	(1.4)	2,610	(1.2)	0.016
Statins	97,089	(44.1)	574,945	(42.0)	0.043	97,066	(44.1)	94,036	(42.8)	0.028
Other lipid-lowering agents	40,985	(18.6)	198,117	(14.5)	0.112	40,968	(18.6)	39,518	(18.0)	0.017
Use of antidiabetic drugs										
Metformin	113,930	(51.8)	757,414	(55.3)	-0.071	113,897	(51.8)	112,259	(51.0)	0.015
α-glucosidase inhibitors	4,980	(2.3)	49,801	(3.6)	-0.081	4,978	(2.3)	4,622	(2.1)	0.011
GLP-1 receptor agonists	1,007	(0.5)	1,530	(0.1)	0.065	982	(0.4)	780	(0.4)	0.015

Insulin	21,708	(9.9)	143,486	(10.5)	-0.020	21,688	(9.9)	19,877	(9.0)	0.028
Meglitinides	857	(0.4)	8,494	(0.6)	-0.033	856	(0.4)	761	(0.3)	0.007
Sulfonylurea	49,873	(22.7)	406,212	(29.7)	-0.160	49,861	(22.7)	49,027	(22.3)	0.009
Thiazolidinediones	14,695	(6.7)	73,395	(5.4)	0.055	14,684	(6.7)	13,939	(6.3)	0.014
Lv of antidiabetic treatment§										
1	154,244	(70.1)	891,149	(65.1)	0.113	154,242	(70.1)	156,210	(71.0)	0.029
2	44,024	(20.0)	334,186	(24.4)		44,010	(20.0)	43,853	(19.9)	
3	21,708	(9.9)	143,486	(10.5)		21,688	(9.9)	19,877	(9.0)	
Healthcare use										
Hospitalizations										
0	175,965	(80.0)	1,040,981	(76.0)	0.112	175,937	(80.0)	178,212	(81.0)	0.029
1-2	39,814	(18.1)	283,101	(20.7)		39,806	(18.1)	38,108	(17.3)	
≥3	4,197	(1.9)	44,739	(3.3)		4,197	(1.9)	3,620	(1.6)	
Number of outpatient visits										
0-2	15,505	(7.0)	97,519	(7.1)	0.056	15,505	(7.0)	15,927	(7.2)	0.010
3-5	20,248	(9.2)	104,679	(7.6)		20,246	(9.2)	20,613	(9.4)	
≥6	184,223	(83.7)	1,166,623	(85.2)		184,189	(83.7)	183,400	(83.4)	

† Data are number (%) unless otherwise specified. All covariates were assessed during the year before cohort entry, including the cohort entry date.

‡ Red-colored number indicates negligible difference (absolute standardized mean difference <0.1)

§ Use of antidiabetic drugs past 1 year: (1) only one antidiabetic drug or without any antidiabetic drug; (2) at least two classes of noninsulin antidiabetic drugs; or (3) at least one insulin treatment as alone or in combination with other antidiabetic drugs.

Abbreviations: ACEI, angiotensin converting enzyme inhibitor; ARB, angiotensin II receptor blocker; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; GLP-1, glucagon-like peptide-1; MRA, mineralocorticoid receptor antagonist; SD, standard deviation; SGLT2i, sodium-glucose co-transporter 2 inhibitor

Table S8. Results of stratified analyses of rate differences and hazard ratios for amputation associated with the use of SGLT2i compared with DPP4i

Analyses	Rate difference (95% CI)	PS-matched HR (95% CI)	*P* for interaction
CVD+/DU+			
Main analysis	-0.76 (-1.42 to -0.09)	0.36 (0.14 to 0.90)	
Aged <65 years	-0.95 (-1.85 to -0.05)	0.23 (0.05 to 1.06)	0.435
Aged ≥65 years	-0.55 (-1.53 to 0.43)	0.50 (0.15 to 1.63)	
Male	-0.94 (-2.14 to 0.26)	0.46 (0.18 to 1.19)	-
Female	-	-	
CVD+/DU-			
Main analysis	-0.34 (-0.68 to -0.00)	0.45 (0.21 to 0.99)	
Aged <65 years	-0.29 (-0.72 to 0.13)	0.50 (0.19 to 1.34)	0.725
Aged ≥65 years	-0.41 (-0.97 to 0.14)	0.38 (0.10 to 1.39)	
Male	-0.47 (-0.94 to -0.00)	0.39 (0.15 to 1.00)	0.565
Female	-0.13 (-0.58 to 0.33)	0.62 (0.15 to 2.62)	
CVD-/DU+			
Main analysis	-0.26 (-0.56 to 0.05)	0.54 (0.26 to 1.12)	
Aged <65 years	-0.37 (-0.72 to -0.02)	0.39 (0.15 to 1.00)	0.213
Aged ≥65 years	0.02 (-0.59 to 0.64)	0.99 (0.30 to 3.24)	
Male	-0.38 (-0.89 to 0.13)	0.54 (0.23 to 1.26)	0.986
Female	-0.14 (-0.46 to 0.18)	0.55 (0.14 to 2.20)	
CVD-/DU-			
Main analysis	-0.26 (-0.39 to -0.13)	0.48 (0.33 to 0.70)	
Aged <65 years	-0.20 (-0.34 to -0.07)	0.55 (0.37 to 0.83)	0.178
Aged ≥65 years	-0.55 (-0.92 to -0.17)	0.26 (0.10 to 0.68)	
Male	-0.34 (-0.54 to -0.14)	0.49 (0.32 to 0.75)	0.843
Female	-0.16 (-0.31 to -0.01)	0.45 (0.21 to 0.98)	
Overall cohort			
Main analysis	-0.22 (-0.33 to -0.11)	0.54 (0.40 to 0.73)	
Aged <65 years	-0.15 (-0.27 to -0.04)	0.63 (0.44 to 0.89)	0.223
Aged ≥65 years	-0.44 (-0.71 to -0.16)	0.41 (0.24 to 0.72)	
Male	-0.36 (-0.54 to -0.19)	0.50 (0.36 to 0.70)	0.309
Female	-0.05 (-0.16 to 0.06)	0.73 (0.38 to 1.41)	

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease. In each cohort, SGLT2i users were propensity score matched to DPP4i users with 1:1 ratio based on all potential confounders at baseline.

Abbreviations: CI, confidence interval; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; HR, hazard ratio; SGLT2i, sodium-glucose cotransporter 2 inhibitor

Table S9. Results of sensitivity analyses of rate differences and adjusted hazard ratios for amputation associated with the use of SGLT2i compared with DPP4i

Analyses	Rate difference (95% CI)	PS-matched HR (95% CI)
CVD+/DU+		
Main analysis	-0.76 (-1.42 to -0.09)	0.36 (0.14 to 0.90)
Intention-to-treat approach	-1.25 (-2.24 to -0.27)	0.35 (0.15 to 0.83)
Grace period of 0 day	-1.19 (-3.23 to 0.84)	0.28 (0.03 to 2.65)
Grace period of 60 days	-0.63 (-1.26 to 0.01)	0.44 (0.20 to 1.01)
Redefined study groups (At least two records for defining baseline status)	-0.76 (-1.59 to 0.06)	0.32 (0.09 to 1.16)
Redefined new users (2-year washout period)	-0.83 (-1.66 to -0.01)	0.37 (0.13 to 1.03)
CVD+/DU-		
Main analysis	-0.34 (-0.68 to -0.00)	0.45 (0.21 to 0.99)
Intention-to-treat approach	-0.52 (-1.03 to -0.01)	0.50 (0.25 to 1.00)
Grace period of 0 day	N/A	N/A
Grace period of 60 days	-0.36 (-0.69 to -0.03)	0.46 (0.23 to 0.93)
Redefined study groups (At least two records for defining baseline status)	-0.54 (-0.94 to -0.14)	0.36 (0.16 to 0.81)
Redefined new users (2-year washout period)	-0.19 (-0.56 to 0.17)	0.61 (0.24 to 1.54)
CVD-/DU+		
Main analysis	-0.26 (-0.56 to 0.05)	0.54 (0.26 to 1.12)
Intention-to-treat approach	-0.41 (-0.83 to 0.00)	0.45 (0.19 to 1.03)
Grace period of 0 day	-0.31 (-1.21 to 0.58)	0.48 (0.04 to 5.34)
Grace period of 60 days	-0.35 (-0.65 to -0.05)	0.47 (0.24 to 0.91)
Redefined study groups (At least two records for defining baseline status)	-0.14 (-0.45 to 0.17)	0.67 (0.30 to 1.48)
Redefined new users (2-year washout period)	-0.40 (-0.76 to -0.04)	0.38 (0.15 to 0.96)
CVD-/DU-		
Main analysis	-0.26 (-0.39 to -0.13)	0.48 (0.33 to 0.70)
Intention-to-treat approach	-0.31 (-0.47 to -0.14)	0.52 (0.36 to 0.74)
Grace period of 0 day	-0.60 (-1.05 to -0.14)	0.33 (0.13 to 0.85)
Grace period of 60 days	-0.26 (-0.39 to -0.14)	0.47 (0.33 to 0.67)
Redefined study groups (At least two records for defining baseline status)	-0.20 (-0.32 to -0.08)	0.55 (0.39 to 0.79)
Redefined new users (2-year washout period)	-0.11 (-0.25 to 0.03)	0.69 (0.44 to 1.09)
Overall cohort		
Main analysis	-0.22 (-0.33 to -0.11)	0.54 (0.40 to 0.73)
Intention-to-treat approach	-0.29 (-0.43 to -0.15)	0.56 (0.42 to 0.75)
Grace period of 0 day	-0.55 (-0.90 to -0.20)	0.32 (0.14 to 0.71)
Grace period of 60 days	-0.24 (-0.34 to -0.14)	0.53 (0.40 to 0.69)
Redefined new users (2-year washout period)	-0.25 (-0.38 to -0.12)	0.52 (0.37 to 0.73)

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease. In each cohort, SGLT2i users were propensity score matched to DPP4i users with 1:1 ratio based on all potential confounders at baseline.

Abbreviations: CI, confidence interval; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; HR, hazard ratio; SGLT2i, sodium-glucose cotransporter 2 inhibitor

Table S10. Results of the hospitalization for heart failure among propensity score matched population

Group	Events / Patients	Person years	Rate per 1,000 PY	Rate difference (95% CI)	PS-matched HR (95% CI)
CVD+/DU+					
SGLT2i	231 / 11,719	13,187	17.52 (15.40 to 19.93)	-2.32 (-5.53 to 0.88)	0.84 (0.71 to 1.00)
DPP4i	292 / 11,719	14,718	19.84 (17.69 to 22.25)	Reference	Reference
CVD+/DU-					
SGLT2i	90 / 26,092	29,651	3.04 (2.47 to 3.73)	-1.78 (-2.76 to -0.80)	0.62 (0.48 to 0.81)
DPP4i	157 / 26,092	32,615	4.81 (4.12 to 5.63)	Reference	Reference
CVD-/DU+					
SGLT2i	74 / 26,894	34,910	2.12 (1.69 to 2.66)	-1.11 (-1.86 to -0.36)	0.64 (0.48 to 0.85)
DPP4i	124 / 26,894	38,381	3.23 (2.71 to 3.85)	Reference	Reference
CVD-/DU-					
SGLT2i	150 / 155,195	163,878	0.92 (0.78 to 1.07)	-0.25 (-0.47 to -0.03)	0.77 (0.63 to 0.95)
DPP4i	201 / 155,195	172,092	1.17 (1.02 to 1.34)	Reference	Reference
Overall cohort					
SGLT2i	545 / 219,940	241,661	2.26 (2.07 to 2.45)	-0.74 (-1.02 to -0.45)	0.74 (0.66 to 0.82)
DPP4i	771 / 219,940	257,567	2.99 (2.79 to 3.21)	Reference	Reference

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease. In each cohort, SGLT2i users were propensity score matched to DPP4i users with 1:1 ratio based on all potential confounders at baseline.

Abbreviations: CI, confidence interval; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; DU, diuretic; HR, hazard ratio; SGLT2i, sodium-glucose cotransporter 2 inhibitor; PS, propensity score; PY, person years

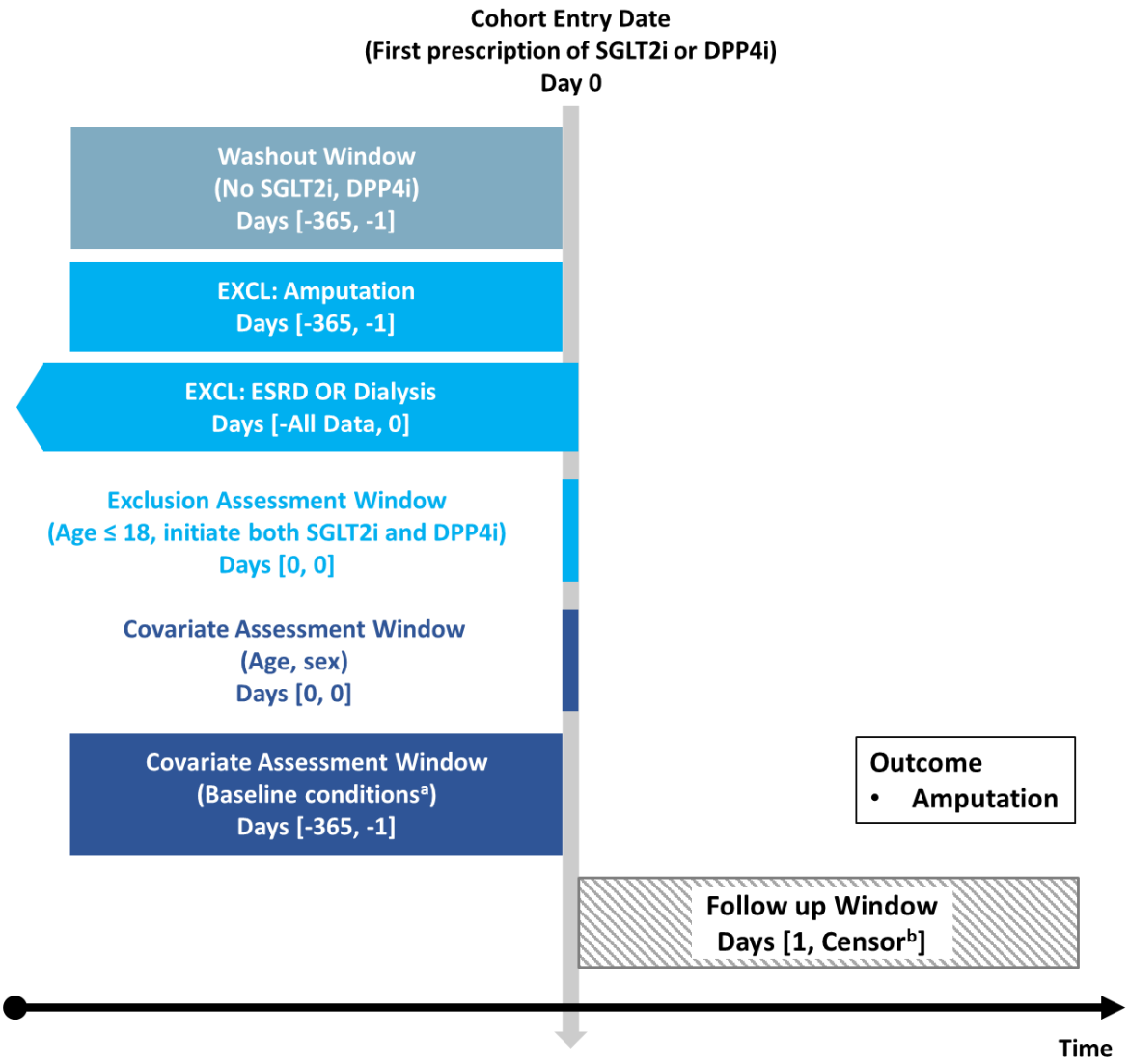


Figure S1. Graphical representation of study design
Note: (a) variables generating propensity score, and baseline status of cardiovascular disease, and diuretic use; (b) censoring criteria: occurrence of outcome, switching within comparison group, discontinuation of index drug, death, end of study period.
Abbreviation: DPP4i, dipeptidyl peptidase-4 inhibitors; ESRD, end-stage renal disease; SGLT2i, sodium-glucose cotransporter 2 inhibitor

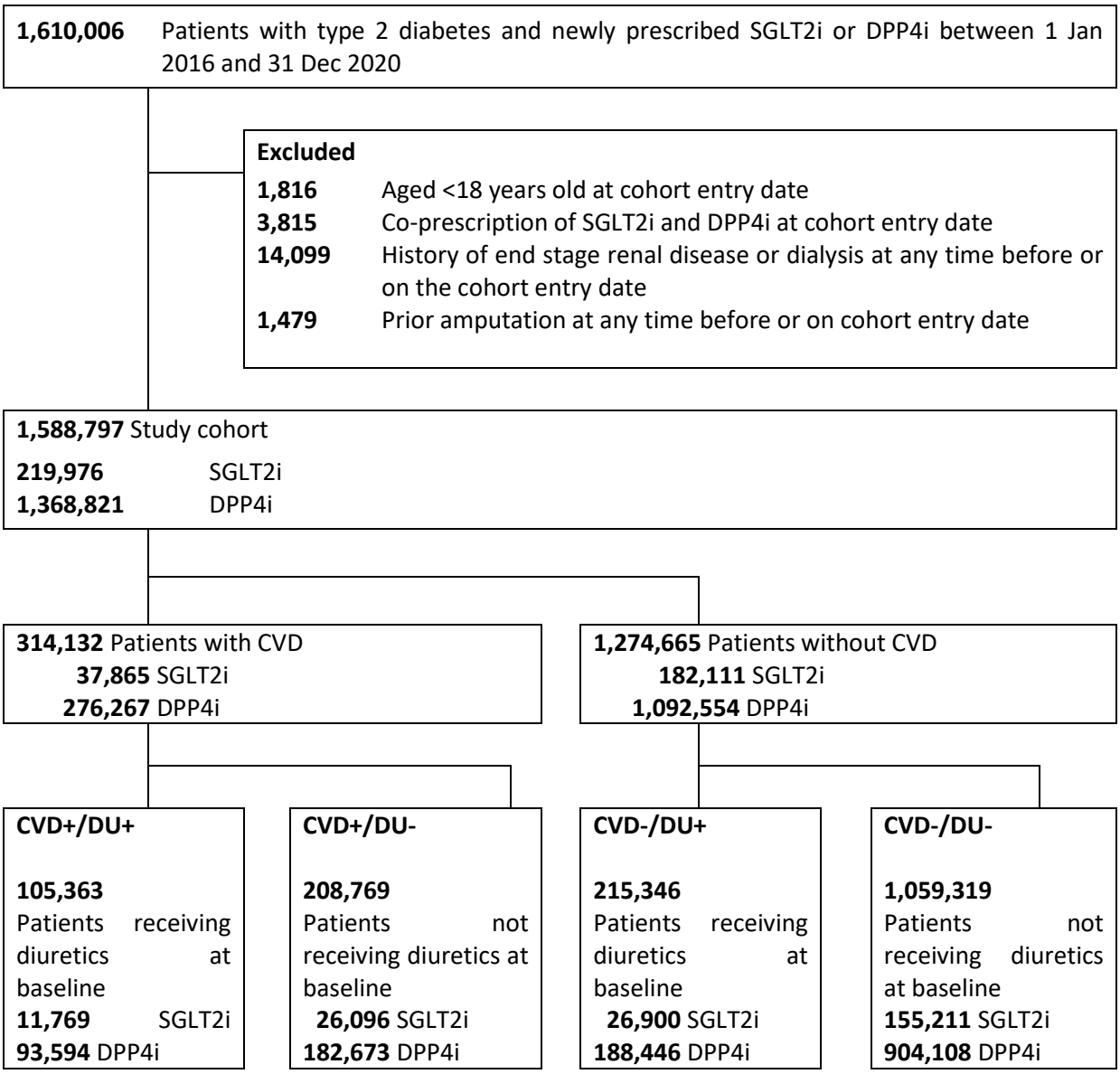
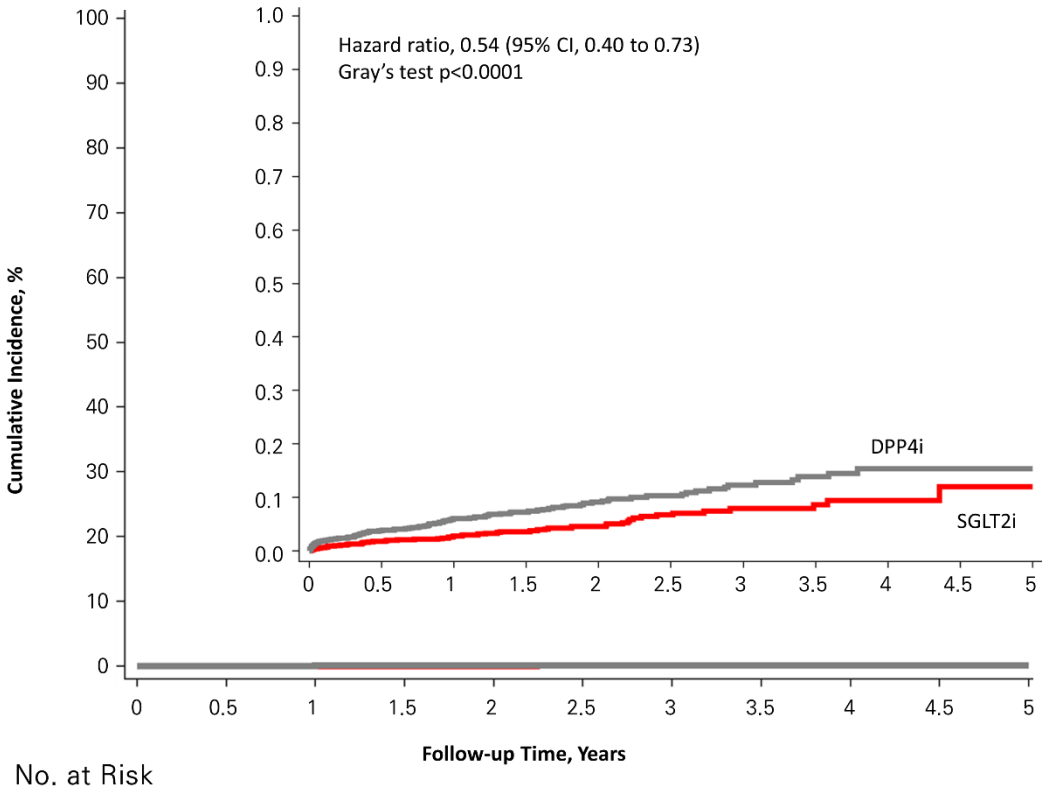


Figure S2. Flow chart describing selection of the study cohort.

Note: Patients were classified into four mutually exclusive groups based on the baseline status in the year prior to cohort entry: CVD+/DU+, patients with cardiovascular disease and diuretic use; CVD+/DU-, patients with cardiovascular disease only (no diuretic use); CVD-/DU+, patients with diuretic use only (no cardiovascular disease); CVD-/DU-, patients without cardiovascular disease and diuretic use. Cardiovascular disease included cerebrovascular disease, coronary artery disease, heart failure, peripheral vascular disease.

Abbreviation: CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitors; DU, diuretic; SGLT2i, sodium-glucose cotransporter 2 inhibitor



No. at Risk

DPP4i	219,940	127,801	87,887	64,275	45,613	34,257	24,097	16,364	9,106	4,747	14
SGLT2i	219,940	124,613	84,465	59,441	41,468	29,820	20,611	13,556	7,254	2,943	9

Figure S3. Cumulative incidence curves after propensity score matching for overall cohort. Abbreviations: DPP4i, dipeptidyl peptidase-4 inhibitor; SGLT2i, sodium-glucose cotransporter 2 inhibitor.